System Card

Final FaceNet verification system | v2.3 | November 2026

System Overview

This project implements a pair-level face verification system that predicts whether two input facial images belong to the same identity. The pipeline loads and preprocesses two images, generates embeddings using a pretrained **InceptionResnetV1 (FaceNet)** model, computes cosine similarity, and applies a validation-selected threshold.

Inference pipeline: Image A + Image B → image loading and resizing → pretrained face embeddings → cosine similarity → threshold decision → CLI score, prediction, confidence margin, and latency.

Intended Use

The project is intended for educational research, coursework, and reproducibility demonstrations. Appropriate uses include measuring verification performance on Labeled Faces in the Wild (LFW) pairs, comparing face-embedding scores, and understanding CPU inference behavior.

Not intended for: production authentication, surveillance, law enforcement, border control, hiring, financial or public-service eligibility decisions, or other high-stakes identity decisions. The model has not undergone production validation.

Data and Model

The evaluation uses LFW, ingested through TensorFlow Datasets. The project deterministically generates training, validation, and test identity partitions and labeled matching/nonmatching image pairs using a fixed seed and configuration. Images are resized to 160 × 160 and scaled before inference with **InceptionResnetV1 pretrained on VGGFace2**. The pretrained model serves as a feature extractor; it was not retrained on this project's LFW data.

LFW is not representative of all real-world populations or imaging conditions. Reliable demographic labels are not available here, so no subgroup fairness results are claimed.

Verification Decision

For each pair, cosine similarity is compared against the operating threshold of **0.30612244897959173**. A similarity at or above the threshold predicts the same identity (1); a lower similarity predicts different identities (0).

The reported **confidence** is the absolute similarity-score distance from this decision boundary; it is a margin, **not a calibrated correctness probability**.

Threshold Calibration

The threshold was selected using **400 balanced validation pairs** (200 matching and 200 nonmatching). A grid of 50 candidates from −1 to 1 was evaluated with FaceNet similarities. The candidate maximizing validation balanced accuracy was 0.30612244897959173, with **96.25% validation balanced accuracy**. A separate verification run reproduced the same selected value.

Final Held-Out Test Evaluation

With the validation-selected threshold fixed and no test-label-based retuning, the final model was evaluated on **4,046 held-out LFW test pairs**: 2,023 matching and 2,023 nonmatching. The model correctly classified **3,828 pairs**.

Metric	Result
Accuracy	94.61%
Balanced accuracy	94.61%
Precision	93.49%
Recall	95.90%
F1 score	94.68%
ROC-AUC	0.9886
Decision threshold	0.30612244897959173

Actual / Predicted	Different (0)	Same (1)
Different (0)	TN: 1,888	FP: 135
Same (1)	FN: 83	TP: 1,940

These values describe discrimination on the evaluated LFW pairs, not expected performance in production or on other datasets. False-positive matches are a particular concern for consequential identity decisions.

Data Separation and Leakage Checks

An independent check of the generated pair datasets found **zero shared identities and zero shared images** between train/validation, train/test, and validation/test splits. The check also found zero inconsistent pair labels and zero incorrect split assignments.

These checks establish separation between the **project-generated LFW pair splits**. They do not establish whether LFW identities also appeared in the external VGGFace2 pretraining data, which was not audited.

Historical Milestone 2 Baselines

Prior to the FaceNet embedding pipeline, raw-pixel experiments achieved **49.50%** balanced accuracy without normalization and **58.75%** with Z-score normalization. They are historical Milestone 2 results, not final-model performance. See M2_report.pdf for that original methodology and analysis.

CPU Performance Profile

Local CPU profiling on an **Intel Core i5-8350U**, using validation pairs, recorded 0.1312 seconds average end-to-end latency per pair and 7.62 pairs/second for the 25-pair run (p95 latency: 0.1639 seconds per pair). Embedding inference dominated runtime; cosine scoring required comparatively little time. See Profiling report.pdf for the original breakdown and batch-size comparisons.

Common Failure Modes

Performance may degrade with poor lighting, shadows, low resolution, blur, strong pose differences, occlusions, off-center cropping, large age gaps between images, look-alike individuals, and faces outside the LFW distribution. Similarity may be high for a different-identity pair or low for a true match; the single operating threshold does not eliminate either error.

Fairness and Misuse Considerations

The model may inherit dataset biases from both LFW and external pretraining. No demographic-stratified validation was performed, and the observed aggregate accuracy is not evidence of equal performance across subgroups. The system should not be used to make decisions affecting legal rights, employment, access, financial status, public safety, or personal freedom.

Operational Constraints

The pipeline accepts two valid LFW-relative image paths and performs pair-level inference. It is not a scalable similarity search service. TensorFlow, PyTorch, the LFW cache, and pretrained model weights increase setup size. Runtime depends on CPU, OS, caching, model warmup, and whether inference is executed inside Docker. The Docker image demonstrates portability rather than production deployment optimization.

Reproducibility

The canonical reproducibility instructions are in **README.md** and **reports/reproducibility_checklist.md**. The final scripts are **scripts/evaluate_final.py**, **scripts/verify_threshold.py**, and **scripts/check_data_leakage.py**. Run from the project root after preparing the LFW dataset and generated pairs.

Verification command	Purpose
python scripts/verify_threshold.py	Repeat validation-only threshold selection
python scripts/check_data_leakage.py	Check identity, image, label and split separation
python scripts/evaluate_final.py	Compute 4,046-pair held-out test metrics

Final documented release: v2.3. The release tag should reference the commit containing these finalized reports and the associated implementation. If reports were changed after the tag was created, verify the tag's commit before claiming they are included in that release.

Related Documentation

See **M2_report.pdf** for original raw-pixel experiments; **Profiling report.pdf** for measured CPU-stage timings; and **reports/reproducibility_checklist.md** for execution commands and expected checks.